Governed Distributed Cognition: A Model of Stable Reasoning in Long-Horizon Human–AI Systems

This version: OSF preprint, DOI 10.17605/OSF.IO/NCRP2

Author: Rishi Sood (ORCID 0009-0008-6479-4061)

Affiliation: Independent Research Collaboration

Corresponding author: rishisood@protonmail.com

Date: March 2026

Keywords: Human–AI collaboration; Distributed cognition; Governed cognition; Long-horizon reasoning; Cognitive systems; AI-assisted reasoning; Artifact-based memory; Cognitive regulation; Error correction; Human–AI systems

Abstract

Large language models are increasingly used in long-horizon human–AI collaboration, where sustained interaction is required to support complex reasoning and decision-making. In such settings, small deviations in interpretation, context, or reasoning can accumulate over time, undermining the stability and coherence of the collaboration.

This paper proposes a model of cognition for such systems, arguing that cognition in long-horizon human–AI interaction is not located within the human or the AI alone. Instead, it emerges as a distributed, governed, and recoverable process across human judgment, AI reasoning, and artifact-based memory. The analysis defines the governed human–AI cognitive system as the object of study and examines how cognition operates within this structure.

The paper makes three central contributions:

First, it conceptualizes cognition as a distributed process that exists at the level of the interaction system rather than within individual components.

Second, it introduces governance as a constitutive element of cognition, demonstrating how constraints, operational protocols, and linguistic control mechanisms regulate the trajectory of reasoning.

Third, it formalizes recoverability as a defining property of cognition, showing how continuous processes of drift detection and repair enable reasoning to remain coherent over extended interaction sequences.

These properties are integrated into a governed cognitive loop, a recurrent process through which reasoning is generated, evaluated, stabilized, and corrected over time. Together, this framework provides a system-level account of how stable cognition can be sustained in long-horizon human–AI collaboration.

The paper contributes to emerging work on human–AI systems by reframing reliability as a property of interaction design and governance rather than model capability alone, and by providing a foundation for the study of cognition in governed human–AI systems.

1. Introduction

Recent advances in large language models have enabled increasingly complex forms of interaction between humans and AI systems. In many settings, this interaction extends beyond short, isolated exchanges into sustained, long-horizon collaboration, where humans and AI systems work together over extended periods to perform analysis, develop ideas, and make decisions. In such environments, the central challenge is not only the capability of the model, but the stability and coherence of reasoning over time.

A growing body of research has examined the reliability of AI systems from multiple perspectives. Work on foundation models has focused on improving model capabilities and alignment through training and evaluation techniques (Bommasani et al., 2021; Ouyang et al., 2022). Research in human–computer interaction has explored how interface design, prompting strategies, and user oversight influence the effectiveness of AI-assisted work (Amershi et al., 2019). At a broader level, studies of distributed cognition have demonstrated that cognitive processes may extend across individuals, tools, and external representations rather than being confined to a single agent (Hutchins, 1995).

While these approaches address important aspects of performance, interaction, and cognition, they do not fully explain how reasoning remains stable and coherent in long-horizon human–AI collaboration. In such settings, interaction unfolds across extended sequences in which context must be preserved, decisions must remain consistent, and deviations in reasoning must be detected and corrected. The probabilistic nature of language models introduces the possibility of drift, inconsistency, and partial reasoning, which can accumulate over time if not actively managed.

Prior work has addressed this challenge by examining the role of governance in stabilizing long-horizon human–AI collaboration. A series of studies (Sood, 2025a–d; 2026a-b) identified the structural components and operational mechanisms that enable collaboration to remain stable over time, including artifact-based memory, operational protocols, linguistic control signals, and repair processes for detecting and correcting conversational drift. This work formalized the concept of a governance architecture in which reliability emerges as a property of the interaction system rather than of the model alone.

This paper builds on a broader research program examining stability, failure, and governance in long-horizon human–AI collaboration (Sood, 2025a–d; 2026a-b).

The model developed in this paper is derived from longitudinal observation of a governed human–AI collaboration system, documented across extended interaction sequences and recurrent failure–repair cycles (Sood, 2025c).

However, while this prior work explains how collaboration remains stable, it does not fully address a deeper question: how should cognition itself be understood within such governed systems? Existing models of cognition either locate reasoning within the human or describe loosely distributed processes across agents and artifacts. Neither perspective fully captures the structured, regulated, and continuously corrected nature of reasoning observed in long-horizon human–AI collaboration.

This paper addresses that gap by examining cognition within a governed human–AI system. The central claim of this paper is that cognition in long-horizon human–AI collaboration is not located within the human or the AI alone. Instead, it emerges as a distributed, governed, and recoverable process across human judgment, AI reasoning, and artifact-based memory. Within this system, reasoning is continuously regulated by governance mechanisms, anchored in persistent artifacts, and stabilized through ongoing processes of drift detection and repair.

This perspective shifts the unit of analysis from individual agents to the interaction system as a whole. Rather than treating cognition as an internal property of either humans or AI systems, the paper conceptualizes cognition as a system-level process that unfolds through the interaction of multiple components operating under conditions of governance.

The paper therefore examines the following research question:

What is the nature of cognition in governed human–AI systems operating over extended time horizons?

2. Object of Study — The Governed Human–AI Cognitive System

To analyze cognition in long-horizon human–AI collaboration, it is necessary to clearly define the object of study. Existing approaches typically focus on individual components of the interaction, such as the human participant, the AI model, or the interface through which they communicate. While each of these perspectives captures an aspect of the interaction, none is sufficient to explain how reasoning unfolds across sustained collaboration.

In long-horizon settings, cognition does not arise from a single agent operating in isolation. Instead, reasoning emerges from the coordinated interaction of multiple components that operate together over time. The human participant contributes judgment, interpretation, and directional control. The AI system generates responses, explores alternative representations, and expands the space of possible reasoning paths. External artifacts preserve context, intermediate results, and prior decisions, enabling continuity across interaction sequences that extend beyond the limits of transient conversational memory.

These components are not loosely connected. Their interaction is structured and regulated through a set of governance mechanisms that constrain how reasoning unfolds. Objectives define the direction of the task, constraints limit the space of acceptable outputs, operational protocols organize the sequence of reasoning steps, and repair processes correct deviations when they occur. As a result, cognition within the system is not only distributed across components but also shaped by the conditions under which those components interact.

The object of study in this paper is therefore defined as the governed human–AI cognitive system. This system consists of four interdependent elements:

- **Human judgment**, which establishes objectives, interprets results, and evaluates the validity of outputs

- **AI reasoning**, which generates, transforms, and expands representations of the problem space
- **Artifact-based memory**, which preserves context, decisions, and intermediate states across interactions
- **Governance mechanisms**, which regulate the interaction through constraints, operational rules, and repair processes

Cognition in this framework is not reducible to any single element. The human participant cannot reconstruct the full reasoning process without the contributions of the AI system and the persistence of external artifacts. Similarly, the AI system does not maintain continuity of reasoning independently across interaction sequences. The cognitive process exists only at the level of the system, emerging from the interaction of its components under governance.

This definition establishes a clear boundary for the analysis that follows. The paper does not examine human cognition in isolation, AI cognition in isolation, or loosely distributed cognition across unregulated systems. Instead, it focuses specifically on cognition as it arises within a governed human–AI system operating over extended time horizons.

The structure of this system is illustrated in **Figure 1**, which shows the interaction between human judgment, AI reasoning, artifact-based memory, and governance mechanisms within a unified cognitive architecture.

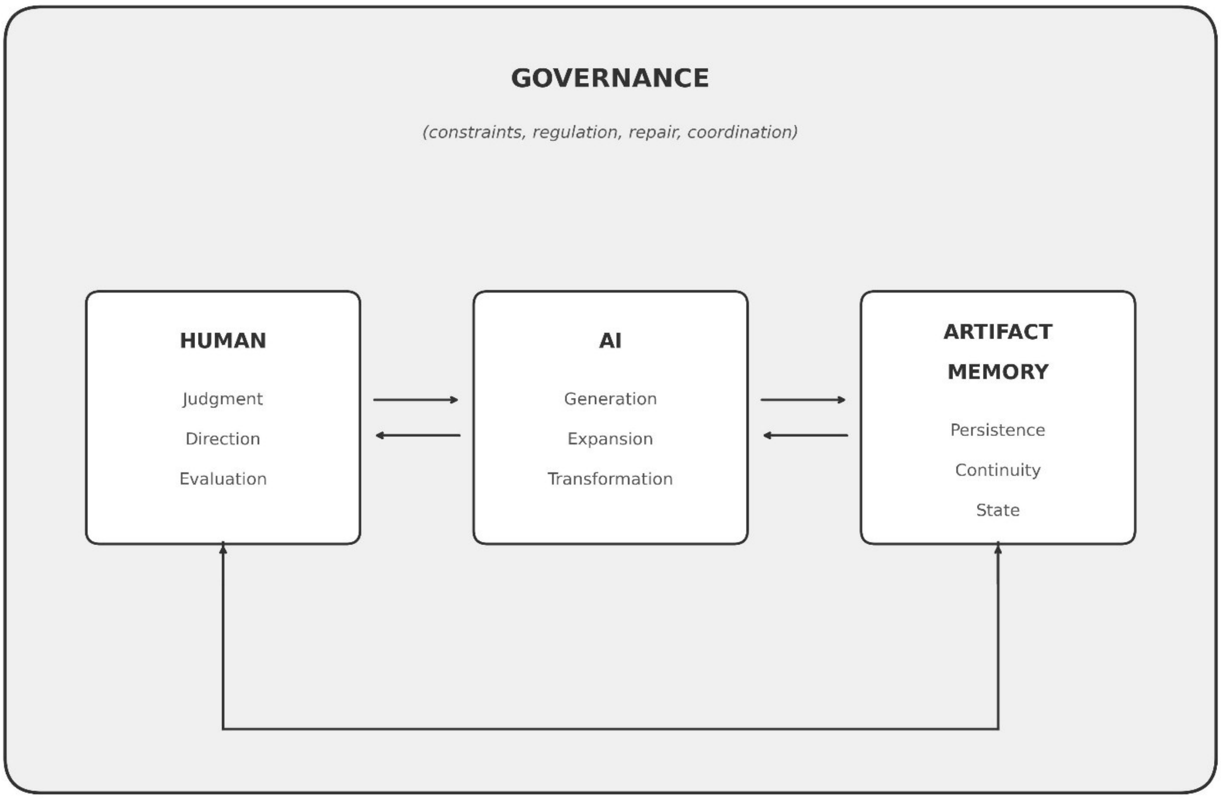

Figure 1. *Structure of the governed human–AI cognitive system. Human judgment defines objectives and evaluates outputs, AI reasoning generates and transforms representations,*

artifact-based memory preserves persistent state across interactions, and governance mechanisms regulate the interaction through constraints, operational protocols, and repair processes. Together, these components form an integrated system in which cognition emerges at the level of the interaction rather than within any single element.

3. Distributed Cognition

Cognition in long-horizon human–AI collaboration does not reside within a single agent. Instead, it emerges from the interaction of multiple components that together constitute the cognitive process. This distributed nature is not merely a matter of collaboration between separate entities, but a structural condition in which no individual component contains the complete reasoning process.

Within the governed human–AI system, each component performs a distinct and partial role. The human participant contributes judgment, interpretation, and directional intent, determining what the system is trying to achieve and evaluating whether intermediate outputs are valid. The AI system generates candidate representations, explores alternative reasoning paths, and transforms information into new forms that expand the space of possible interpretations. Artifact-based memory preserves context, intermediate results, and prior decisions, enabling continuity across interaction sequences that extend beyond the limits of transient conversational memory.

These components operate interdependently. The human relies on AI-generated outputs to explore possibilities that would be difficult to produce independently. The AI system depends on human direction and evaluation to constrain its generative space. Artifact-based memory provides a persistent substrate that allows both human and AI contributions to be accumulated, referenced, and refined over time. None of these components can independently sustain the full reasoning process; cognition exists only through their coordinated interaction.

This perspective departs from traditional models that locate cognition within the human mind, as well as from approaches that treat AI systems as autonomous reasoning agents. It also extends prior work on distributed cognition, which demonstrates that cognitive processes may be shared across individuals, tools, and external representations (Hutchins, 1995). While such frameworks establish that cognition can be distributed, they typically describe systems in which coordination is loosely organized and contingent on situational factors. However, these frameworks do not specify mechanisms through which distributed cognition is actively regulated and made recoverable over extended time horizons.

In contrast, the system examined in this paper exhibits a more structured form of distribution. The components of the system are not loosely connected but operate within a persistent and organized interaction in which prior state is preserved, referenced, and actively used in subsequent reasoning. As a result, cognition is not only distributed across components but also sustained across time through the accumulation and stabilization of intermediate states.

The distributed nature of cognition in this system implies a shift in the unit of analysis. Rather than attributing reasoning to the human or the AI system, cognition must be understood at the level of the interaction system itself. The cognitive process cannot be reconstructed from any single component in isolation; it exists only as a property of the system formed by their interaction.

This distributed structure forms the foundation for the subsequent claims of the paper. While cognition is distributed across components, it is not unstructured. The interaction between components is regulated through governance mechanisms, and the resulting cognitive process is stabilized through continuous detection and correction of deviations. The following sections examine how governance and recoverability transform distributed cognitive activity into a stable and sustained form of reasoning over extended time horizons.

4. Governance as Cognitive Regulation

While cognition in the governed human–AI system is distributed across multiple components, it does not operate as an unconstrained or purely emergent process. Instead, the interaction between components is structured and regulated in ways that determine how reasoning unfolds over time. In this system, governance is not an external layer applied to an otherwise independent cognitive process. It is a constitutive element of cognition itself, determining the conditions under which reasoning is generated, evaluated, and maintained over time.

Governance operates through a set of mechanisms that constrain and direct the interaction between human judgment, AI reasoning, and artifact-based memory. Objectives define the direction of the task and establish what the system is attempting to achieve. Constraints limit the space of acceptable outputs, preventing uncontrolled expansion of the reasoning process. Operational protocols structure how reasoning progresses, organizing interaction into sequences that promote completeness and coherence. Linguistic control signals further regulate behavior by shaping how instructions are interpreted and executed during interaction.

Through these mechanisms, governance determines which reasoning paths are explored, how intermediate outputs are evaluated, and how consistency with prior state is maintained. The cognitive process is therefore not free-form. It is continuously shaped by conditions that influence the generation, interpretation, and validation of outputs at each stage of interaction.

This role of governance distinguishes the system examined in this paper from traditional accounts of distributed cognition. While existing frameworks demonstrate that cognitive processes may be distributed across agents and artifacts, they typically do not specify mechanisms that actively regulate how such processes unfold over extended time horizons. Coordination in these systems is often contingent and loosely organized, rather than systematically constrained.

In contrast, the governed human–AI system exhibits a structured form of interaction in which regulation is persistent and explicit. The presence of governance mechanisms ensures that reasoning remains aligned with objectives, consistent with prior state, and responsive to detected deviations. As a result, governance does not merely support cognition; it actively constitutes the conditions under which cognition is possible.

This perspective implies that cognition in such systems cannot be understood independently of its regulatory structure. The stability, directionality, and coherence of reasoning emerge from the presence of governance mechanisms that continuously shape the interaction between components. Without these mechanisms, the distributed elements of the system would produce fragmented and inconsistent reasoning, lacking the continuity required for sustained intellectual work.

Governance therefore functions as a form of cognitive regulation. It does not sit outside the cognitive process but operates within it, structuring how reasoning progresses and ensuring that the interaction remains aligned over time. The result is a form of cognition that is not only distributed, but also actively organized and constrained as it unfolds.

5. Recoverable Cognition

Cognition in governed human–AI systems is not only distributed and regulated but fundamentally recoverable. This property distinguishes it from both traditional models of human cognition and existing accounts of distributed cognitive systems.

In conventional accounts of cognition, errors are typically treated as disruptions to reasoning. Misinterpretations, incorrect assumptions, or logical inconsistencies may degrade the quality of thought and require external correction. While humans are capable of revising their reasoning, such correction is not structurally embedded within the cognitive process itself. Similarly, in standard AI interaction, incorrect or incomplete outputs may be addressed through subsequent prompts, but these corrections are ad hoc and not integrated into a persistent system of reasoning.

In contrast, governed human–AI systems incorporate explicit mechanisms through which deviations in reasoning are identified and corrected as part of the cognitive process itself. These deviations may include contextual drift, inconsistencies with prior state, incomplete reasoning, or misalignment with task objectives. Rather than allowing such deviations to accumulate, the system detects them during ongoing interaction.

Recovery is achieved through structured processes of detection, diagnosis, and repair. First, deviations are identified through comparison between current outputs and the governing conditions of the system, including objectives, constraints, and prior artifacts. Second, the nature of the deviation is diagnosed, distinguishing between different types of failure such as misinterpretation, omission, or inconsistency. Third, corrective action is applied, which may involve re-anchoring the reasoning process in prior state, clarifying intent, or restructuring the current line of reasoning.

These processes are not occasional interventions but continuous functions of the system. Error detection and repair are integrated into the operation of cognition, ensuring that local deviations do not propagate into systemic instability. As a result, cognition in this framework is not a linear progression of reasoning steps but a regulated process that includes its own mechanisms of correction.

This recoverability transforms the role of error within cognition. Rather than being treated solely as a source of disruption, error becomes a point of intervention through which reasoning is stabilized and redirected. The presence of structured repair mechanisms allows the system to maintain coherence over extended interaction sequences despite the inherent variability of generative AI systems.

The recoverable nature of cognition is grounded in longitudinal empirical observation of failure and repair in governed human–AI collaboration, where recurrent patterns of drift, misalignment, and correction were systematically documented (Sood, 2025c).

For example, repeated interaction sequences show cases in which incomplete reasoning or contextual drift is identified and corrected through explicit repair steps, restoring alignment with prior state (Sood, 2025c).

Analysis of failure patterns demonstrates that instability arises not from the presence of errors alone, but from the accumulation of uncorrected deviations over time. Conversely, systems that incorporate explicit mechanisms for detecting and correcting such deviations are able to maintain coherence across extended sequences of interaction.

This perspective reframes stability in cognitive systems. Stability is not achieved by eliminating error, but by embedding the capacity for recovery within the cognitive process itself. Cognition becomes a self-correcting process in which deviations are continuously identified and resolved, enabling reasoning to persist over time without degrading into inconsistency.

Recoverability therefore represents a defining property of cognition in governed human–AI systems. It ensures that distributed and regulated cognitive activity can remain coherent across extended time horizons, transforming a potentially fragile process into one that is structurally capable of maintaining alignment and continuity.

6. The Governed Cognitive Loop

Cognition in governed human–AI systems does not unfold as a linear sequence of reasoning steps. Instead, it operates as a recurrent, regulated process in which distributed components interact continuously to produce, evaluate, stabilize, and extend reasoning over time. This process can be understood as a governed cognitive loop that integrates generation, evaluation, and repair within a single operational structure.

The loop begins with intent formation, in which the human participant defines the objective, constraints, and criteria that guide the reasoning process. This step establishes the direction of cognition and determines the conditions under which subsequent outputs will be evaluated.

Following this, generative expansion occurs. The AI system produces candidate representations, interpretations, or solution paths, expanding the space of possible reasoning trajectories. This stage increases the dimensionality of the cognitive process by introducing alternative structures and perspectives that would be difficult to generate through human reasoning alone.

The outputs of this expansion are then stabilized through state anchoring. Key elements of the reasoning process are externalized into artifact-based memory, preserving intermediate results, decisions, and contextual information. This anchoring ensures continuity across interaction sequences and prevents the loss of structure that would otherwise occur in transient conversational exchanges.

Once anchored, the system performs evaluation and selection. Generated outputs are assessed against the governing conditions of the system, including objectives, constraints, and prior state. This stage determines which elements of the expanded reasoning space are retained, refined, or discarded, shaping the trajectory of the cognitive process.

Crucially, the loop incorporates drift detection as an explicit stage of operation. Deviations from intended reasoning—such as inconsistencies, incomplete steps, or misalignment with objectives—are identified during ongoing evaluation. These deviations represent points at which the cognitive process risks diverging from its intended trajectory.

When such deviations are detected, the system initiates repair and realignment. The reasoning process is corrected by re-anchoring in prior artifacts, clarifying intent, or restructuring the current line of reasoning. As established in the previous section, this repair process is not external to cognition but intrinsic to its operation.

The loop concludes with continuation, in which the corrected and stabilized state becomes the basis for subsequent reasoning. The system proceeds into the next cycle, extending the cognitive process while maintaining alignment with its governing conditions.

This sequence—intent formation, generative expansion, state anchoring, evaluation, drift detection, repair, and continuation—constitutes a recurring cycle through which cognition is sustained over extended time horizons. Each iteration of the loop not only advances reasoning but also stabilizes it, ensuring that deviations are corrected before they accumulate into systemic instability.

The governed cognitive loop is illustrated in **Figure 2**, which represents the cyclical structure through which distributed, governed, and recoverable cognition is enacted in long-horizon human–AI collaboration.

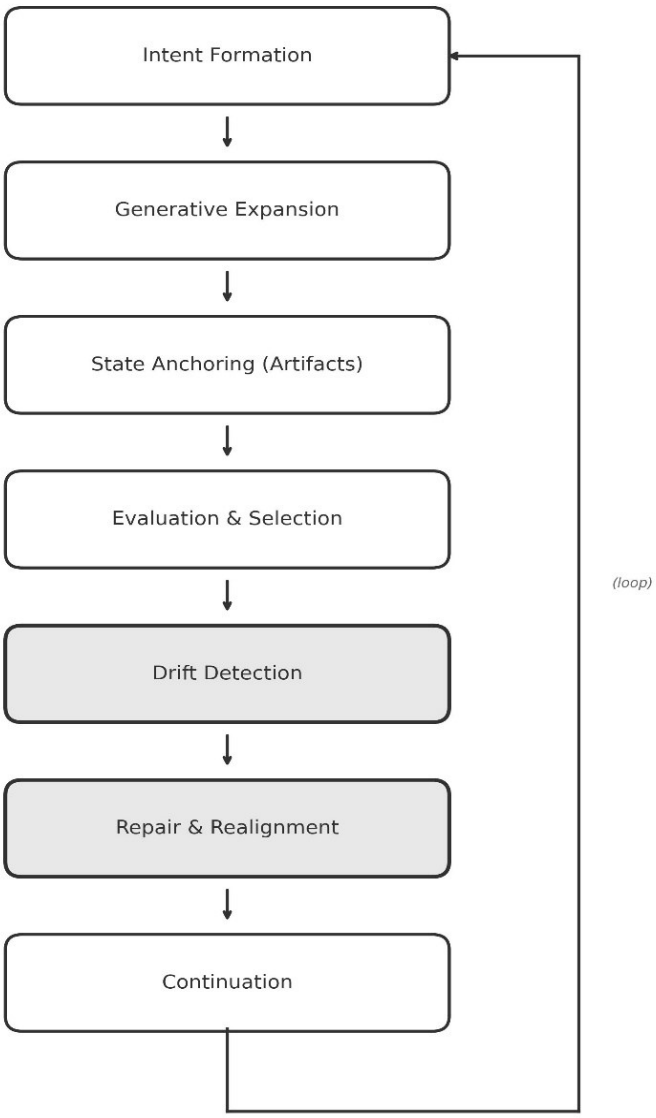

Figure 2. *The governed cognitive loop. Cognition unfolds as a recurring cycle of intent formation, generative expansion, state anchoring, evaluation, drift detection, repair, and continuation. This loop integrates generation, regulation, and correction within a single process, enabling reasoning to remain coherent and aligned across extended interaction sequences.*

This formulation provides an operational model of cognition in governed human–AI systems. Rather than viewing reasoning as a unidirectional progression, the loop structure emphasizes that cognition advances through repeated cycles of expansion, stabilization, evaluation, and correction. The integration of governance and recoverability within this cycle ensures that the cognitive process remains both adaptive and stable over time.

7. Implications and Limits

The model of cognition proposed in this paper does not aim to provide a general theory of cognition. Instead, it identifies a specific class of cognitive systems that emerge under conditions of governed human–AI interaction. The claims advanced here are therefore bounded by the structure and conditions of the system under study.

One immediate implication of this framework is a shift in how reliability in human–AI collaboration is understood. Rather than treating reliability as a property of the AI model or as a function of improved prompting techniques, the analysis suggests that stability arises from the structure of the interaction system itself. In particular, the combination of distributed cognition, governance mechanisms, and embedded repair processes enables reasoning to remain coherent over extended time horizons despite the variability of generative models.

This perspective has implications for the design of AI-assisted work systems. If cognition in long-horizon collaboration is distributed, governed, and recoverable, then improving performance requires attention not only to model capabilities but also to the structure of interaction. Systems that incorporate persistent artifact-based memory, explicit constraints, and mechanisms for detecting and correcting deviations are more likely to sustain coherent reasoning than systems that rely solely on ad hoc interaction.

At the same time, the framework highlights the limitations of unstructured human–AI interaction. In the absence of governance mechanisms, distributed cognitive activity does not produce stable reasoning over time. Instead, interaction is susceptible to drift, inconsistency, and loss of context, leading to the accumulation of errors that degrade the quality of outputs. This suggests that improvements in model capability alone are insufficient to address the challenges of long-horizon collaboration without corresponding advances in interaction design.

The model also has implications for how cognition is studied in human–AI systems. By shifting the unit of analysis from individual agents to the interaction system, the framework suggests that cognitive properties such as stability, coherence, and recoverability should be understood as system-level phenomena. This perspective may inform future empirical and theoretical work on hybrid human–AI cognition, particularly in settings where sustained reasoning is required.

However, the scope of the present work is limited in several important respects. First, the model is derived from the analysis of a single, highly governed human–AI collaboration. While this provides detailed insight into system dynamics, it does not establish the generality of the framework across different users, tasks, or organizational contexts. Further research is required to determine whether the mechanisms identified here apply more broadly.

Second, the framework assumes the presence of explicit governance structures, including artifact-based memory, operational protocols, and repair mechanisms. Systems that lack these features may not exhibit the same cognitive properties. As a result, the model should not be interpreted as describing all forms of human–AI interaction, but only those that meet the structural conditions outlined in this paper.

Third, the analysis focuses on the structure of cognition rather than its efficiency or optimality. While the framework explains how stable reasoning is maintained, it does not address questions of performance trade-offs, computational cost, or scalability. These considerations are important for practical implementation but fall outside the scope of the present work.

Finally, the framework does not attempt to resolve broader philosophical questions about the nature of cognition or agency. Its purpose is more limited: to describe how cognition operates within a specific class of governed human–AI systems, based on empirical observation and system-level analysis.

Taken together, these limitations define the boundaries of the contribution. The paper does not claim that all cognition is distributed, governed, and recoverable. Rather, it demonstrates that under specific conditions—those of governed human–AI interaction—cognition takes on these properties and can be sustained over extended time horizons.

8. Conclusion

This paper examined the nature of cognition in long-horizon human–AI collaboration and proposed a model of cognition as a distributed, governed, and recoverable process. By defining the governed human–AI cognitive system as the object of study, the analysis shifted the focus from individual agents to the interaction system within which reasoning unfolds.

The framework developed in this paper establishes that cognition in such systems does not reside within the human or the AI alone. Instead, it emerges from the coordinated interaction of human judgment, AI reasoning, and artifact-based memory, operating under conditions of governance. Within this structure, reasoning is continuously shaped by constraints, stabilized through externalized state, and maintained through ongoing processes of detection and repair.

The introduction of governance as a constitutive component of cognition extends existing accounts of distributed cognition by demonstrating that cognitive processes can be actively regulated and stabilized over extended time horizons. The concept of recoverable cognition further reframes the role of error, showing that stability is achieved not by eliminating deviations, but by embedding mechanisms that detect and correct them as part of the cognitive process itself.

The governed cognitive loop provides an operational model of how these properties are enacted in practice. Through recurring cycles of generation, evaluation, anchoring, and repair, the system maintains coherence while continuing to evolve over time. This loop structure integrates distribution, regulation, and recoverability into a single process that enables sustained reasoning in long-horizon interaction.

Taken together, these contributions provide a system-level account of cognition in governed human–AI collaboration. The paper does not propose a general theory of cognition, but identifies a specific class of cognitive systems in which reasoning is distributed across components, shaped by governance, and sustained through continuous recovery mechanisms.

As human–AI interaction increasingly moves toward sustained and complex forms of collaboration, understanding cognition at the level of the interaction system becomes essential. The model presented here provides a foundation for further investigation into such systems, including their design, evaluation, and extension to broader contexts.

The model is derived from observed interaction dynamics within a governed human–AI system, rather than from purely theoretical construction, and is intended to provide a foundation for further empirical and system-level investigation.

Acknowledgements

The authors thank the open-source and academic AI community for foundational work on context engineering, governance frameworks, and reproducible tooling. The authors also acknowledge independent researchers whose public discussions of prompt and context management helped inform some of the control concepts formalised here. Special thanks are extended to “Mahdi” (AI system used through ChatGPT) for analytical assistance, structural reasoning, and drafting support during the development of this paper.

Author Contributions

Rishi Sood designed the research, maintained the authoritative artefacts, and conducted longitudinal validation. He takes full responsibility for the claims and interpretation in this paper. “Mahdi” (AI system used through ChatGPT) assisted by implementing and documenting control procedures, generating analytical summaries, and proposing draft text under the author's direction. All final wording, framing, and conclusions were reviewed and approved by the author.

Funding

This work received no external funding and was conducted as part of an independent research collaboration.

Competing Interests

The authors declare no commercial or financial relationships that could be construed as a potential conflict of interest.

References

Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., … Horvitz, E. (2019).
Guidelines for human-AI interaction. Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems, 1–13.

Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., … Liang, P. (2021).
On the opportunities and risks of foundation models. arXiv preprint arXiv:2108.07258.

Hutchins, E. (1995).
Cognition in the Wild. Cambridge, MA: MIT Press.

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., … Lowe, R. (2022).
Training language models to follow instructions with human feedback. Advances in Neural Information Processing Systems (NeurIPS), 35.

Sood, R. (2025a).
Context-engineered human–AI collaboration for long-horizon tasks: A case study in governance, canonical numerics, and execution control. OSF preprint. https://doi.org/10.17605/OSF.IO/VMK7Y

Sood, R. (2025b).
The Lean Collaboration Operating System (LC-OS): A practical framework for long-term human–AI work. OSF preprint. https://doi.org/10.17605/OSF.IO/695AF

Sood, R. (2025c).
Failure and repair in long-horizon human–AI collaboration: A transparent tracing case study. OSF/Zenodo preprint. https://doi.org/10.17605/OSF.IO/Z7AQ8

Sood, R. (2025d).
The living framework: Living with a governed human–AI dyad. OSF/Zenodo preprint. https://doi.org/10.17605/OSF.IO/ER4YT

Sood, R. (2026a).
Control Without Code: Linguistic Governance in Long-Horizon Human–AI Collaboration. OSF/Zenodo preprint. https://doi.org/10.17605/OSF.IO/HJKWG

Sood, R. (2026b).
Governance Architecture for Reliable Long-Horizon Human-AI Collaboration. OSF/Zenodo preprint. https://doi.org/10.17605/OSF.IO/N2A78